1(a)(i)	(halogens can) remove electron(s) (from a reactant)	**1**

Examiner report missing
9701_w25_qp_24 Question 1

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**
1(c)(ii)	(+)4	**1**
1(c)(iii)	*Due to an issue with question 1(c)(iii), the question has been removed from the question paper.*	**N/A**

Examiner report missing
9701_w25_qp_23 Question 1

1(a)(i)	25 protons 30 neutrons 25 electrons	**1**
1(c)(i)	Mn / MnO_2 **AND** it gains electrons	**1**
1(c)(ii)	(+)4	**1**
1(c)(iii)	*Due to an issue with question 1(c)(iii), the question has been removed from the question paper.*	**N/A**

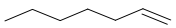
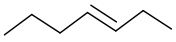
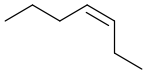
Examiner report missing
9701_w25_qp_22 Question 1

1(a)(i)	24 protons 28 neutrons 24 electrons	**1**
1(c)(i)	Cr / $Cr_2O_7^{2-}$ **AND** it gains electrons	**1**
1(c)(ii)	(+)4	**1**
1(c)(iii)	*Due to an issue with question 1(c)(iii), the question has been removed from the question paper.*	**N/A**

Examiner report missing
9701_w25_qp_21 Question 1

2(a)(i)	non-linear	**1**
2(b)(i)	*In any order:* **M1** (solution / liquid) (red) to colourless **M2** yellow solid appears	**2**
2(b)(ii)	oxidation number of S in SCl_2 = (+) 2 / (+) II oxidation number of S in H_2SO_3 = (+) 4 / (+) IV	**2**

Examiner report missing
9701_s25_qp_24 Question 2

5(a)	**M1**  **M2**  **OR** 	**2**

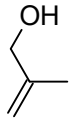
5(d)(i)	oxidising agent	**1**

Examiner report missing
9701_s25_qp_24 Question 5

2(a)	Na_2O	**1**

2(c)(i)	SO_3	**1**

Examiner report missing
9701_s25_qp_23 Question 2

5(a)(i)	C_2H_3O	**1**
5(b)(ii)		**1**
5(b)(iii)	reducing agent	**1**

Examiner report missing
9701_s25_qp_23 Question 5

4(a)(i)	$LiAlH_4$	**1**
4(c)(i)	sp^3	**1**
4(c)(ii)	hydrogen / H_2	**1**
4(c)(iii)	(It / Na) reduces **A** / **B** / **C** / the alcohol (group) **OR** (it / Na behaves as) reducing agent **OR** (it / Na) donates electrons to **A** / **B** / **C** / the alcohol (group)	**1**

Examiner report missing
9701_s25_qp_22 Question 4

1(a)(i)	sodium – metallic **AND** sodium oxide – ionic	**1**

1(c)	**M1** NaCl +1 **AND** PCl_5 + 5 **M2** Na has only 1 valence (shell) electrons **AND** P has 5	**2**

1(d)	*All three for two marks:* • false • comparison or statement of oxidation number • correct link to melting point from their comparison or statement	**2**

Examiner report missing
9701_s25_qp_21 Question 1

4(a)	$C_3H_4O_3$	**1**
4(b)(i)	$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5}$ mol	**1**
4(b)(iii)	oxidising agent	**1**

Examiner report missing
9701_s25_qp_21 Question 4

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**

Examiner report missing
9701_m25_qp_22 Question 1

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(a)(iii)	green colour disappears	**1**
2(b)(ii)	ultraviolet	**1**
2(b)(iii)	$C_2H_6 + Cl^{\bullet} \rightarrow C_2H_5^{\bullet} + HCl$	**1**
	$C_2H_5^{\bullet} + Cl_2 \rightarrow C_2H_5Cl + Cl^{\bullet}$	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

(iii) Candidate answers that were awarded credit stated that 'white fumes' would be observed although strictly this occurs only when moisture is present.

(iv) Many candidate answers linked the decreasing thermal stability of hydrogen halides to weaker intermolecular forces of attraction. Some candidates only mentioned bond length and did not link the decrease in thermal stability to the decrease in bond strength of the H-Hal bond.

(b) (i) Many candidate answers stated that it was an 'unpaired electron or a lone pair of electrons'. The requirement for a species to be specified, that contained 'one or more unpaired electrons', was often omitted.

(ii) This question was well answered.

(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

(c) (i) Many candidates recognised that NaOH was required to convert chlorine to NaClO, but the requirement for the NaOH to be in aqueous solution and the conditions to be 'cold' were regularly omitted.

(ii) The definition of disproportionation was well answered by many candidates.

(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

3(a)(i)	(natural =)lightning	**1**
	(man-made =)internal combustion engines	**1**
3(a)(v)	brown fumes given off	**1**
3(b)(i)	(+)5 / V	**1**
3(b)(ii)	aluminium / A*l*	**1**

Question 3

(a) (i) Many candidate answers referred to lightning as responsible for one natural occurrence of NO_2. A common incorrect answer was acid rain. Fewer candidate answers described the main source of man-made nitrogen oxides as the internal combustion engine. Lots of candidates incorrectly identified the source as catalytic converters or 'car exhaust' as the origin of nitrogen oxides.

(ii) Very few candidates gave appropriate species in the equation and if they did these equations were not always balanced correctly. Many missed the significance of the word 'direct' in the question and attempted to give an equation showing the role of NO_2 as a catalyst in the production of sulfur-based acid rain.

(iii) Candidates who were familiar with this area of the syllabus described how PAN forms from NO_2. Common incorrect descriptions included acid rain, water, oxygen, carbon monoxide, carbon dioxide or ozone reacting with NO_2.

(iv) Correctly balanced equations were common. Typical errors included incorrect formula of nitric acid or calcium nitrate or production of H_2 rather than H_2O.

(v) Some candidate answers described the production of a brown gas. Others focused on the production of a white solid even though the original compound is also a white solid.

(b) (i) Occasionally incorrect oxidation states were seen; –1 and +3 were common incorrect answers.

(ii) The majority of candidates identified aluminium as the correct species that is oxidised. The weakest answers described one of the two products of the reaction.

(iii) Excellent answers gave a full description of the formation of a dative bond due to the lone pair on the nitrogen atom accepting a proton. Weaker candidate descriptions only referred to the ability of ammonia to form a dative bond.

(iv) Identification of an appropriate shape for $Al(OH)_4^-$ was present in a significant proportion of answers.

(c) (i) Inclusion of state symbols for ionisation energy equations are essential. Some equations showed addition of an electron, other equations were incorrectly balanced.

(ii) A general increase in ionisation energy was shown by most candidates. Many answers showed a larger increase in energy for one of the ionisation processes, with the exact ionisation process often incorrectly identified. Weaker answers confused the trend in ionisation energy across a period with the trend of successive ionisation energies.

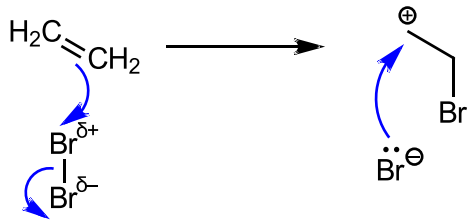
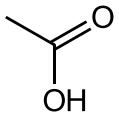
(d) Phosphorus was sometimes identified incorrectly with ionic bonding or as an ionic structure.

(e) Use of the term 'simple' to identify the structure of solids made up of discrete molecules was not well known and the structure of solid nitrogen was often referred to as 'giant'.

(f) (i) This question highlighted many misconceptions about hybridisation, sigma and pi bonds. These include

- a triple bond involves formation of two sigma bonds and one pi bond
- triple bonds are made when sigma and pi overlap
- hybridisation in P_2 identified as sp^2 or sp^3
- in P_2 overlap of s orbitals creates a sigma bond
- formation of sigma bonds by side-on overlap of orbitals and pi bonds by head-on overlap of orbitals.

(ii) Many candidate responses referred to the appropriate difference in bond strength. Some answers were incomplete, with no comparison of reactivity given. Some candidates misinterpreted the question and explained why they had different bond energy values.

4(a)	 **pt 1** dipole on Br_2 **pt 2** curly arrow from C=C bond to a $Br^{\delta+}$ **pt 3** curly arrow from Br–Br bond to the other Br **pt 4** correct intermediate **pt 5** curly arrow from lone pair on Br^- to C^+ **5•** = 3 marks. **4•** = 2 marks. **3•** = 2 marks. **2•** = 1 marks. **1•** = 0 marks.	**3**
4(d)(i)	structural / positional	**1**
4(e)(i)	**M1** identify **H** =  **M2** (broad) absorption within the range 3600–2500 cm^{-1} so O-H (bond) O-H (bond) is equivalent to OH bond **M3** pt 1 and pt 2 •✓ **OR** pt 1 and pt 3 •✓ •**pt 1** absorption within the range 1670–1750 cm^{-1} so C=O (bond) •**pt 2** absorption within the range 1040–1300 cm^{-1} so C-O (bond) •**pt 3** (M^+ at m/e = 60 so it has) molecular mass / Mr = 60	
4(e)(ii)	oxidising agent	**1**

Question 4

(a) Well-rehearsed candidate answers included the accurate placement of curly arrows, dipoles and the correct intermediate. Weaker answers often had ambiguous placement of curly arrows, inappropriate use of partial charges, incorrect dipole on Br_2 and incorrect intermediates.

(b) Correct calculations, sometimes with an appropriate Hess' cycle in their working, were seen. Incorrect answers were calculated by those candidates that did not fully understand or use the definition of enthalpy change of formation when applying Hess's law.

(c) (i) The product **A** was correctly shown in many answers. Candidates need to show correct connectivity of the O-H group to the carbon backbone when structures are drawn.
The naming of **B** presented problems for some candidates. Incorrect names often described the incorrect position of Br on the carbon backbone or gave a name which described the wrong number of carbon atoms in the backbone.

(ii) Unambiguous structures were seen in a significant number of candidate answers.

(iii) This question proved to be extremely demanding. All the information provided in the details of Fig. 5.1 needed to be considered, including the three products as well as the reaction conditions.

(d) (i) Correct identification of the type of isomerism was given in a significant number of candidate answers. Chain isomerism was a common incorrect answer even though the two molecules were made up of only two carbon atoms.

(ii) Many candidate answers stated that **F** was a carbonyl compound; this term describes the functional group present in the molecule and is common to both homologous series; aldehyde and ketone.

(iii) Many candidate answers gave appropriate observations for the behaviour of **F** with 2,4-DNPH reagent and Tollen's reagent. A much smaller number of answers identified a positive reaction with appropriate observation for the reaction with alkaline I_2(aq).

(e) (i) This question proved to be demanding. The strongest candidates gave full explanations of their interpretation of the infrared spectrum by relating relevant wavenumber values at absorption troughs to appropriate bond types and used the molecular ion peak to deduce the molecular mass of **H**. Weaker responses quoted key wavenumber values at troughs in the infrared spectrum and restated that the molecular ion peak in the mass spectrum was at m/e = 60 without explaining what this information actually provides about structure **H**.

(ii) The correct role of reagent **G** was given in many candidate answers. Catalyst was a common incorrect answer.

2(a)(i)	chlorine = yellow-green bromine = orange / brown / red iodine = silver grey / black	**1**
2(b)(ii)	ultraviolet	**1**
2(c)(ii)	a reaction where a species / same species is both oxidised and reduced simultaneously / in the same reaction	**1**

Question 2

(a) (i) The colours of the Group 17 elements, chlorine, bromine and iodine were not well known with common errors being 'yellow' for chlorine and 'purple' for iodine. Many candidate answers stated that the colours became darker down the group which was insufficient.

(ii) Most candidate answers correctly stated that the oxidising strength of the elements chlorine, bromine and iodine decreased down the group.

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(iii) Few candidates were awarded full credit here. Incorrect responses showed that candidates were not clear what a propagation step involved and gave initiation or termination steps instead. Some included ions instead of radicals or gave H radical as an incorrect response.

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(iii) This equation was well known.

(iv) This question was well answered by many candidates. Several candidates lost marks for giving additional incorrect observations for example: a brown gas of NO_2 is also formed. Those candidates who described how the Cl^- ion was formed, prior to its reaction with Ag^+ (in $AgNO_3$) lost credit by failing to appreciate that the $CHCl_3$ was initially hydrolysed to form Cl^-. Many stated that the Cl^- was contained in $CHCl_3$, and not a product from its hydrolysis.

1(a)(i)	**M1** (N_2 molecules /molecule(s) of N_2) have a strong triple (covalent) bond **M2** (N_2 molecules /molecule(s) of N_2) are non-polar	**2**
1(b)(i)	(+)III	**1**
1(d)(i)	(+)IV	**1**

Question 1

This question assessed inorganic chemistry, with some aspects of physical chemistry. This tested principally in the AO1 category. Candidates showed some facility of recall but were equally divided in response when asked to apply their knowledge to unfamiliar circumstances.

(a)(i) This standard recall question could be answered with easy reference to syllabus points.

(ii) With reference to the syllabus, orbital overlap can be directly quoted; knowledge of the bonding in N_2 is also straightforward, so the number of σ and π bonds should be readily accessible.

(b)(i) Many candidates answered this question well.

(ii) This question gave rise to many variations of answer. Candidates are advised to be brief with their answers: often, correct responses were contradicted by (incorrect) explanations of valence electrons.

(c)(i) Many candidates answered this question well.

(ii) The reaction of P_4O_{10} was better known than that of Al_2O_3. The latter is not less common but requires better honed balancing skills if constructed rather than written in recall.

(d)(i) Many candidates answered this question well.

(ii) Many candidates were able to identify the type of reaction that occurs.

1(a)	<table><tr><td>halogen</td><td>colour at 293 K</td></tr><tr><td>chlorine</td><td>green</td></tr><tr><td>bromine</td><td>brown</td></tr><tr><td>iodine</td><td>(dark) grey</td></tr></table> All three correct for one mark	**1**
1(c)(i)	$Br_2 + 2I^- \rightarrow 2Br^- + I_2$	**1**
1(c)(ii)	(Br_2 / bromine is an) oxidising agent **AND** removes electron(s) from iodide (ions) I^- **OR** increases the oxidation number of I^- / iodide (ion)	**1**

Question 1

(a) The colour of the halogens (at 293 K) was not often answered correctly by candidates. Common errors included chlorine described as yellow and iodine as a purple gas. Several candidates gave the colours of silver halides.

(b) The volatility of the halogens chlorine, bromine and iodine is associated specifically with the increased strength of the instantaneous dipole-induced dipole forces between the molecules due to the increase in the number of electrons in these molecules. Many candidates correctly stated that the volatility decreases from chlorine to bromine to iodine. This trend was then incorrectly attributed or linked to the electronegativity variations of the elements or the attraction between the nucleus and the outer electrons.

(c) (i) The majority of candidates used the full formula equation rather than ionic for the reaction of bromine with sodium iodide. Many answers included the spectator ion Na^+ in the equation.

(ii) It was recognised by several candidates that bromine, when reacted with sodium iodide, behaved as an oxidising agent to form iodine and a bromide. Many explanations then incorrectly referred to a change in oxidation number, or movement of electrons, in iodine instead of the iodide ions.

(d) (i) Many candidates gave a correct equation for the reaction of NaBr with concentrated sulfuric acid. Common errors were $NaSO_4$, for sodium sulfate(VI).

(ii) Candidates often identified NaCl as the salt that produced the largest percentage of hydrogen halide. The explanations frequently referred to the reactivity of the sodium halides instead of describing the trend in the hydrogen halides or halide ions as reducing agents.

The most common incorrect answers identified ‘sodium iodide’ producing the largest percentage yield of hydrogen halide as ‘it is the most reactive’.

2(a)(i)	hydrolysis	**1**

2(c)(i)			**1**

	oxidation number in Mg_3N_2
magnesium	(+)II / 2(+)
nitrogen	–III/ 3 –

Both correct for one mark

Question 2

(a) (i) This question was well answered.

(ii) The majority of answers gave a formula for the different Period 3 element when the question asked for a name. Common errors included phosphorus chloride, omitting the oxidation number, and phosphorus pentachloride, which is a solid.

(b) (i) The dot-and-cross diagram of SCl_2 was well answered. A common error included annotating the electrons around the S atom as 'x' in contradiction of the rubric.

(ii) The VSEPR theory was not well known when applied to a central S atom with two bonding pairs and two lone pairs of electrons in the outer shell. The majority of candidates focused on the two bonding pairs in SCl_2, stating that the shape was linear with a bond angle of 180°.

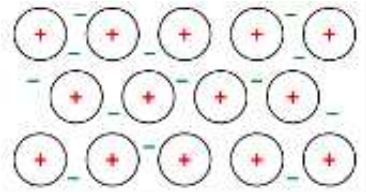
(c) (i) This question was well answered.

(ii) Most candidates gave a correct equation for the reaction of magnesium nitride with water. The most common error was an incorrect formula for magnesium nitride and magnesium hydroxide given as MgOH or MgO.

(iii) The marks were awarded to a minority of candidates. Very few candidates stated that hydroxide ions were present either from the dissociation of $Mg(OH)_2$ in solution, or from a reaction between ammonia and water.

(d) (i) The empirical formula of boron nitride was well answered with a common error of Br used to represent boron.

(ii) Many candidates gave the correct answer of graphite that has a structure similar to that of boron nitride. SiO_2 and graphene were common incorrect answers.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**
1(c)(i)	Bi_2O_3 (+)3 / III **AND** $NaBiO_3$ (+)5 / V	**1**
1(c)(ii)	Bi_2O_3	**1**
1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

2(a)		**3**

formula of chloride	● $NaCl$	● $MgCl_2$	● $SiCl_4$
structure	giant	● giant	● simple
bonding of chloride	● ionic	● ionic	covalent
pH of solution formed	● 7	6.2	● 1–4

2(b)(i)	disproportionation	**1**

Question 2

This was a holistic question based around the chemistry of chlorine and chlorides. Candidates are recommended to give brief answers to one-mark questions and to draw diagrams clearly.

(a) Common errors were found in the formulae of the chlorides, the structure of $SiCl_4$ and in the pH of a $NaCl$ solution (neutral so pH 7; candidates often seemed to think that: $NaCl + H_2O \rightarrow HCl + NaOH$).

(b) (i) Many variant and phonetic spellings of 'disproportionation' were observed.

(ii) There was an issue with this question in that water was not included as one of the products of the reaction, so full marks were awarded to all candidates to make sure no candidates were disadvantaged. This has been corrected in the published version of the paper.

(iii) This equation was rarely constructed correctly: where more valid attempts were made, common errors were either that $NaClO$ was given as the oxidation product or that the correct stoichiometric coefficients were omitted.

(iv) This was answered well by most candidates, although a number offered effervescence as an observation.

(c) (i) Octahedral was the most common answer for PCl_6^-, although commonly hexagonal or octagonal were seen (and not credited). A broad range of answers were given for PCl_4^+, with 'trigonal pyramidal' (missing the bi- prefix) and 'bipyramidal' (missing the base shape) among them.

(ii) **J** was often correctly identified as H_2O/water. Hydration or displacement were often given instead of hydrolysis or substitution for the type of reaction.

(d) (i) Some very loosely drawn diagrams were unable to be given full credit: curly arrows should come from a clear origin (the middle of a bond, or from pair of electrons, rather than a negative charge) and, in cases such as this, the arrowhead needs to have both 'blades' and point to a specific atom or the middle of a bond.

Many candidates also confused partial charges/dipoles with full formal charges.

(ii) Candidates were required to reference heating the ethanolic NaOH to gain full credit.

(iii) Many answers retained the C=C or arranged the 3C monomer linearly with no branching. These were both common and recurring errors in questions of this type.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(c)	**C*l*** / **chlorine** is both oxidised and reduced	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(b)(i)	enthalpy change when one mole of a compound / substance is formed	**1**
	from its constituent elements in their standard states	**1**

3(b)(iii)	P (in H_3PO_3) is (both) oxidised and reduced (simultaneously)	**1**
	P (is oxidised) from (+)3 / (+)III → (+)5 / (+)V AND (reduced to) –3 / –III	**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(c)	**C*l*** / **chlorine** is both oxidised and reduced	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

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1(a)(i)	giant metallic	**1**
1(d)(i)	$2Cu^{2+} + 4I^{-} \rightarrow 2CuI + I_2$	**1**
1(d)(ii)	Cu^{2+} OR $CuSO_4$ **AND** copper (species / ion) has gained / taken electron(s) from iodide (ion)	**1**

Question 1

(a) (i) A high proportion of candidates knew that copper is metallic. Many candidates omitted that the lattice is giant.

(ii) Diagrams showing the metallic bonding as an array of positive metal ions surrounded by delocalised electrons were varied, with many either omitting or not correctly labelling the positive ions and delocalised electrons.

(b) (i) Some candidates incorrectly referred to a mole of ^{12}C atoms or just an atom of ^{12}C.

(ii) Many correct answers were seen. Some candidates omitted the word 'average'.

(iii) The question was quite well answered. The major error included the use of 63 and 65 as the masses of the copper isotopes instead of the data given in the question.

(c) Very few candidates recognised that the ion with an abundance of 23% and an *m/e* of 31.5 was that of $^{63}Cu^{2+}$.

(d) (i) The equation was not well answered overall. Few candidates recognised that it was a redox reaction. Many answers contained electrons on one side of the equation and/or were completely unbalanced.

(ii) The identification of Cu^{2+} as the oxidising reagent in this reaction was generally not well known. Many candidates answered with just 'copper'.

(iii) The full electronic configuration of Cu^{2+} was frequently incorrect with the $4s^2$ electrons still present in many answers.

2(a)(i)	it / aluminium oxide is insoluble (in water) **OR** it / aluminium oxide does not react (with water)	**1**

2(c)(i)	A*l* (+)3 / III **AND** P (+)5 / V	**1**

Question 2

(a) (i) Many candidates knew that aluminium objects covered in a layer of aluminium oxide do not react with cold water because the aluminium oxide is insoluble in water and does not react with the water.

(ii) The equation, with state symbols, was quite well answered, with many candidates recognising that the product was aqueous sodium aluminate.

(iii) Some good answers were seen.

(b) Some candidates gave the formulae of nitrates of elements that are not located in Period 3.

(c) (i) Several candidates correctly stated the oxidation number of Al in $AlCl_3$. The answers for that of P in PCl_5 were varied.

(ii) Many candidates knew that Al and P had different numbers of electrons available to use. Many omitted where these electrons are located.

(iii) It was well recognised that PCl_5 was the required chloride of phosphorus and that HCl is a product of this reaction. Few candidates gave the correct product containing phosphorus. Many answers contained an oxide of phosphorus and/or an incorrect phosphoric acid.

(iv) Some candidates knew the reaction is a hydrolysis reaction with other candidates correctly describing it as a substitution reaction.

(v) Some good answers were seen.

2(a)(i)	<table><tr><td>element</td><td>Na</td><td>Mg</td><td>Al</td><td>Si</td><td>P</td></tr><tr><td>maximum oxidation number of elements in chlorides</td><td>(+)1</td><td>(+)2</td><td>(+)3</td><td>(+)4</td><td>(+)5</td></tr></table>All correct for one mark	**1**

Question 2

(a) (i) The maximum oxidation numbers of the Period 3 chlorides were well known.

(ii) The majority of responses were correct. Some candidates produced responses that were vague and incomplete. Occasionally, explanations were incorrectly given in terms of changes to the nuclear charge.

(b) (i) The majority of candidates identified the correct formula of the silicon chloride and gave an appropriate balanced equation for the reaction.

(ii) A significant proportion of candidates incorrectly described the pH of the solution as greater than or equal to 7.

(c) (i) The equation to represent the reaction for phosphorus(V) chloride with water was less well known than the equation for **(b)(ii)**. The production of phosphorus oxide rather than phosphoric acid was a common incorrect answer. Other equations which showed appropriate species were not correctly balanced.

(ii) Many candidates gave appropriate suggestions for the pH of the solution produced in **(c)(i)**.

(d) (i) Equations using C*l* rather than C*l*$_2$ and production of O_2 with HC*l* were common incorrect answers.

(ii) A small proportion of candidates identified all the correct species and produced a balanced equation. Some candidates found the equation for the reaction of chlorine with hot NaOH(aq) to be demanding.

(e) (i) A wide range of incorrect species were seen in the answers; many of which were not ions.

(ii) Many correct equations representing this reaction were seen. Occasionally, equations gave incorrect multiples for the correct species which included the decomposition of some hydrogen peroxide without the use of sodium chlorate(I).

(iii) A significant proportion of candidates used the appropriate data to calculate the number of moles of sodium chlorate(I) used in a single titration. Difficulties arose in attempts to work out the concentration in the original sample, **W**. Occasionally, concentrations were left in mol dm^{-3} rather than converting this into g dm^{-3}, as detailed in the question.

1(a)	<table><tr><td></td><td>nucleon number</td><td>number of neutrons</td><td>number of electrons</td></tr><tr><td>Tellurium-130</td><td>130</td><td>78</td><td>52</td></tr></table>	**3**
1(e)	**M1** 6 bonding pairs (of electrons and 0 lone pairs) **OR** coordination number = 6 **OR** $x = 6$ **M2** bonding pairs repel (equally)	**2**
1(f)(i)	$TeF_x + xH_2O \rightarrow Te(OH)_x + xHF$ **OR** $TeF_6 + 6H_2O \rightarrow Te(OH)_6 + 6HF$	**1**

Question 1

(a) This was generally well answered. Some candidates were not familiar with the term nucleon number, often interpreting this as neutron number.

(b) Many candidates answered this question correctly. Common errors included not specifying **1**s and giving 5p as the answer.

(c) Candidates generally answered this well. State symbols were not always included, especially in Te^{+}.

(d) (i) The majority of candidates correctly identified two factors. Many candidates incorrectly stated that the shielding effect remains constant. An 'increase in nuclear charge' was a common incorrect response that negated other descriptions.

(ii) Most candidates recognised a general increase in successive ionisation energies, with a majority of those able to indicate the most significant jump from the 6th to 7th ionisation energy. Many answers incorrectly included a graph with a dip, most often in energy at the $6/7^{th}$ ionisation energy.

(e) Candidates who correctly referenced spin-pair interactions gained credit. Few candidates could use the VSEPR theory correctly.

(f) (i) This was generally not answered well. Many responses included a mixture of numbers and '*x*'s. A small proportion of candidates gave a correctly balanced equation.

(ii) Many candidates correctly identified this as a 'hydrolysis' reaction. A common answer was 'displacement', which was not enough to gain credit.

3(a)	**M1** *appearance* (colourless / yellow) liquid **M2** *structure and bonding* simple / molecular **AND** covalent	**2**
3(c)(i)	hot aqueous sodium hydroxide / hot NaOH(aq)	**1**
3(c)(ii)	oxidation number / state of chlorine / C*l* increases / goes (from 0) to +5 **AND** decreases / goes (from 0) to –1	**1**

Question 3

(a) Very few candidates could state the appearance of $SiCl_4$ at room temperature; more were able to state the correct structure and bonding. The most common incorrect response was giant covalent.

(b) Most candidates correctly referred to the appearance and pH. Candidates often did not refer to the relevant observation and struggled to differentiate the behaviours of the two chlorides.

(c) (i) A large number of candidates stated the correct reagent. Many left out aqueous or dilute as the conditions.

(ii) Many of the candidates defined disproportionation; a small proportion were able to refer to the correct oxidation numbers. Several referred incorrectly to the +1 oxidation number for chlorine.

(d) (i) This was well answered by most candidates.

(ii) Most candidates correctly answered the initiation step. Many candidates found the propagation and termination steps more challenging.

(e) This was well answered by most students.

(f) Many candidates were able to give the correct formulae at m/e = 164.Very few candidates gave both of the correct options at m/e = 166.